Human vs. AI Text Classification: A Comprehensive Study Using Machine Learning and Deep Learning Approaches

Huzaifa Nasir[0009-0000-1482-3268]

National University of Computer and Emerging Sciences,
Islamabad, Pakistan
nasirhuzaifa95@gmail.com

Abstract. The proliferation of AI-generated text poses significant challenges in distinguishing human-written content from machine-generated text. This paper presents a comprehensive study on binary text classification to differentiate between human and AI-generated content using a combination of traditional machine learning and deep learning approaches. We leverage the HC3 (Human-ChatGPT Comparison Corpus) dataset comprising 52,452 text samples and develop a robust classification system employing multiple techniques including Logistic Regression, Random Forest, Support Vector Machines (SVM), XGBoost, Bidirectional LSTM with Attention, and BERT-based models. Our feature engineering approach combines TF-IDF representations with 15 handcrafted linguistic features. Through extensive experimentation and ensemble methods, we achieve an impressive F1-score of 99.59% using a weighted ensemble approach, demonstrating the effectiveness of combining multiple classifiers. The study provides insights into feature importance, model performance comparison, and the potential of ensemble methods in text classification tasks.

Keywords: Text Classification · AI-Generated Text · Deep Learning · Ensemble Methods · Natural Language Processing · BERT · Machine Learning

1 Introduction

The advent of large language models (LLMs) such as GPT-3, GPT-4, and ChatGPT has revolutionized natural language generation, enabling the creation of highly coherent and contextually relevant text. While these advancements have numerous beneficial applications, they also raise concerns regarding misinformation, academic integrity, and content authenticity. The ability to accurately distinguish between human-written and AI-generated text has become increasingly important for various stakeholders, including educators, content moderators, and platform administrators.

This paper addresses the binary classification problem of identifying whether a given text sample is written by a human or generated by an AI system. We present a comprehensive study that explores multiple approaches, from traditional machine learning methods to state-of-the-art deep learning architectures, and demonstrates the effectiveness of ensemble techniques in achieving superior classification performance.

1.1 Motivation

The motivation for this research stems from several critical needs:

- **Academic Integrity:** Educational institutions require tools to detect AI-generated submissions in assignments and examinations.
- **Content Authenticity:** Online platforms need mechanisms to identify synthetic content and maintain authenticity.
- **Misinformation Prevention:** Automated detection can help identify potentially misleading AI-generated content.
- **Research Contribution:** Understanding the distinguishing characteristics between human and AI-generated text advances NLP research.

1.2 Contributions

Our main contributions are:

1. A comprehensive comparative analysis of six different classification approaches including traditional ML and deep learning methods.
2. A hybrid feature engineering strategy combining TF-IDF with handcrafted linguistic features totaling 5,015 features.
3. Extensive evaluation of four ensemble methods: Hard Voting, Soft Voting, Weighted Average, and Stacking.
4. Detailed performance analysis achieving 99.59% F1-score with the weighted ensemble approach.
5. Publicly available implementation and detailed experimental methodology for reproducibility.

2 Related Work

The problem of distinguishing AI-generated text from human-written content has gained significant attention in recent years. Early approaches focused on statistical features and traditional machine learning methods [2]. With the emergence of transformer-based models, researchers have explored both detection and generation perspectives.

2.1 Detection Approaches

Statistical Methods: Early work by Gehrmann et al. utilized perplexity-based features and statistical analysis to detect machine-generated text. These methods rely on the observation that language models tend to produce text with different statistical properties compared to human writing.

Machine Learning Classifiers: Traditional ML approaches using features such as n-grams, POS tags, and syntactic patterns have shown promising results. Studies have demonstrated that ensemble methods combining multiple weak learners can achieve robust performance.

Deep Learning Approaches: Recent work has leveraged pre-trained language models like BERT and RoBERTa for detection tasks. Fine-tuning these models on human vs. AI text datasets has achieved state-of-the-art results in various benchmarks.

2.2 Datasets

The HC3 dataset [1] provides a comprehensive collection of human and ChatGPT-generated responses across multiple domains including finance, medicine, open QA, and Reddit ELI5. This dataset enables controlled experimentation and fair comparison across different approaches.

3 Methodology

3.1 Dataset Description

We utilize the HC3 (Human-ChatGPT Comparison Corpus) dataset, which contains text samples from multiple domains. The dataset statistics are presented in Table 1.

Table 1: Dataset Statistics

Attribute	**Value**
Total Samples	52,452
Training Samples	36,716 (70%)
Validation Samples	7,868 (15%)
Test Samples	7,868 (15%)
Number of Classes	2 (Human, AI)
Class Distribution	Balanced (50-50)
Minimum Text Length	50 characters
Random State	42

The dataset encompasses diverse domains including:

- **Open QA:** General question-answering pairs

- **Finance:** Financial domain-specific content
- **Medicine:** Medical and healthcare-related text
- **Reddit ELI5:** Explanations for complex topics
- **Wikipedia (CS/AI):** Computer science and AI articles

3.2 Data Preprocessing

The preprocessing pipeline consists of several critical steps to ensure data quality and consistency:

Text Cleaning:

- Removal of special characters and excessive whitespace
- Normalization of Unicode characters
- Lowercasing (for TF-IDF features)
- Preservation of sentence structure (for linguistic features)

Quality Filtering:

- Minimum length requirement: 50 characters
- Removal of duplicates
- Validation of label consistency

Data Splitting: The dataset is split using stratified sampling to maintain class balance:

$$\text{Split Ratio} = \{70\% \text{ train}, 15\% \text{ validation}, 15\% \text{ test}\} \tag{1}$$

3.3 Feature Engineering

Our feature engineering approach combines two complementary strategies:

TF-IDF Features We employ Term Frequency-Inverse Document Frequency (TF-IDF) vectorization to capture lexical patterns. The TF-IDF score for a term t in document d is computed as:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t) \tag{2}$$

where:

$$\text{TF}(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}} \tag{3}$$

$$\text{IDF}(t) = \log \frac{N}{|\{d \in D : t \in d\}|} \tag{4}$$

Configuration parameters:

- Maximum features: 5,000
- N-gram range: (1, 2) - unigrams and bigrams
- Minimum document frequency: 2
- Maximum document frequency: 0.95

This results in 5,000 TF-IDF features capturing vocabulary distribution patterns.

Handcrafted Linguistic Features We extract 15 linguistic features designed to capture stylistic and structural differences between human and AI-generated text:

Table 2: Handcrafted Linguistic Features

Feature	Description
Text Length	Total number of characters
Word Count	Total number of words
Avg Word Length	Mean length of words
Sentence Count	Number of sentences
Avg Sentence Length	Mean words per sentence
Lexical Diversity	Type-Token Ratio (TTR)
Stopword Ratio	Proportion of stopwords
Punctuation Ratio	Proportion of punctuation marks
Capital Letter Ratio	Proportion of uppercase letters
Digit Ratio	Proportion of numeric characters
Unique Words Ratio	Proportion of unique words
Long Words Ratio	Proportion of words $>$ 6 characters
Short Words Ratio	Proportion of words $\leq$ 3 characters
Question Marks	Count of question marks
Exclamation Marks	Count of exclamation marks

The Type-Token Ratio (TTR) measures lexical diversity:

$$\text{TTR} = \frac{\text{Number of unique words}}{\text{Total number of words}} \tag{5}$$

The final feature vector combines both representations:

$$\mathbf{x} = [\mathbf{x}_{\text{TF-IDF}}, \mathbf{x}_{\text{linguistic}}] \in \mathbb{R}^{5015} \tag{6}$$

3.4 Classification Models

We implement and evaluate six different classification approaches, divided into traditional machine learning and deep learning categories.

Traditional Machine Learning Models **1. Logistic Regression**

Logistic Regression models the probability of class membership using the logistic function:

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^T\mathbf{x}+b)}} \tag{7}$$

The model is trained by minimizing the binary cross-entropy loss with L2 regularization:

$$\mathcal{L}(\mathbf{w}) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)] + \lambda||\mathbf{w}||_2^2 \tag{8}$$

Configuration: $C = 1.0$ (inverse regularization strength), solver='lbfgs', max_iter=1000

2. Random Forest

Random Forest is an ensemble of decision trees trained on bootstrap samples:

$$\hat{y} = \text{mode}\{\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_T\} \tag{9}$$

where T is the number of trees and $\hat{y}_t$ is the prediction from tree t.

Configuration: n_estimators=200, max_depth=None, min_samples_split=2, random_state=42

3. Support Vector Machine (SVM)

SVM finds the optimal hyperplane that maximizes the margin between classes:

$$\min_{\mathbf{w},b} \frac{1}{2}||\mathbf{w}||^2 + C\sum_{i=1}^{N}\xi_i \tag{10}$$

subject to:

$$y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \tag{11}$$

We use the RBF (Radial Basis Function) kernel:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma||\mathbf{x}_i - \mathbf{x}_j||^2) \tag{12}$$

Configuration: kernel='rbf', $C = 1.0$, gamma='scale', probability=True

4. XGBoost

XGBoost implements gradient boosting with regularization:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(\mathbf{x}_i) \tag{13}$$

The objective function at iteration t is:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{N} l(y_i, \hat{y}_i^{(t)}) + \sum_{k=1}^{t}\Omega(f_k) \tag{14}$$

where $\Omega(f) = \gamma T + \frac{1}{2}\lambda||\mathbf{w}||^2$ is the regularization term.

Configuration: n_estimators=200, learning_rate=0.1, max_depth=7, subsample=0.8

Deep Learning Models 1. Bidirectional LSTM with Attention

The BiLSTM architecture processes sequences in both forward and backward directions:

$$\overrightarrow{\mathbf{h}_t} = \text{LSTM}_{\text{forward}}(\mathbf{x}_t, \overrightarrow{\mathbf{h}_{t-1}}) \tag{15}$$

$$\overleftarrow{\mathbf{h}}_t = \text{LSTM}_{\text{backward}}(\mathbf{x}_t, \overleftarrow{\mathbf{h}_{t-1}}) \tag{16}$$

The bidirectional hidden state is:

$$\mathbf{h}_t = [\overrightarrow{\mathbf{h}_t}, \overleftarrow{\mathbf{h}_t}] \tag{17}$$

The attention mechanism computes context-aware weights:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)}, \quad e_t = \mathbf{v}^T \tanh(\mathbf{W}\mathbf{h}_t) \tag{18}$$

$$\mathbf{c} = \sum_{t=1}^{T} \alpha_t \mathbf{h}_t \tag{19}$$

Architecture specifications:

- Embedding dimension: 128
- LSTM units: 64 (per direction, 128 total)
- Dropout: 0.5
- Attention mechanism: Multiplicative attention
- Output: Binary classification via sigmoid

Training configuration:

- Optimizer: Adam ($\beta_1 = 0.9, \beta_2 = 0.999$)
- Learning rate: 0.001
- Batch size: 64
- Epochs: 10 (early stopped at 4)
- Loss function: Binary cross-entropy

2. BERT (Bidirectional Encoder Representations from Transformers)

BERT utilizes the transformer architecture with multi-head self-attention:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{20}$$

Multi-head attention combines multiple attention mechanisms:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\mathbf{W}^O \tag{21}$$

where:

$$\text{head}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V) \tag{22}$$

We fine-tune the pre-trained 'bert-base-uncased' model:

- Model: bert-base-uncased
- Hidden size: 768
- Attention heads: 12

- Layers: 12
- Parameters: 110M
- Max sequence length: 128 tokens

Fine-tuning configuration:

- Optimizer: AdamW with weight decay
- Learning rate: 2e-5 with linear warmup
- Batch size: 16
- Epochs: 2
- Warmup steps: 500

3.5 Ensemble Methods

We implement four ensemble strategies to combine predictions from multiple models:

Hard Voting Hard voting uses majority voting among classifiers:

$$\hat{y} = \text{mode}\{\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_M\} \tag{23}$$

where M is the number of base models.

Soft Voting Soft voting averages predicted probabilities:

$$\hat{y} = \arg\max_c \frac{1}{M} \sum_{m=1}^{M} P_m(y = c|\mathbf{x}) \tag{24}$$

Weighted Average Weighted average assigns different importance to each model:

$$\hat{y} = \arg\max_c \sum_{m=1}^{M} w_m \cdot P_m(y = c|\mathbf{x}) \tag{25}$$

subject to:

$$\sum_{m=1}^{M} w_m = 1, \quad w_m \geq 0 \tag{26}$$

Our optimized weights based on validation performance:

- XGBoost: 0.25
- Logistic Regression: 0.20
- BERT: 0.20
- SVM: 0.15
- Random Forest: 0.10
- BiLSTM: 0.10

Stacking Stacking trains a meta-learner on base model predictions:

$$\mathbf{z}_i = [P_1(y|\mathbf{x}_i), P_2(y|\mathbf{x}_i), \ldots, P_M(y|\mathbf{x}_i)] \tag{27}$$

$$\hat{y}_i = f_{\text{meta}}(\mathbf{z}_i) \tag{28}$$

We use Logistic Regression as the meta-learner with 5-fold cross-validation to generate out-of-fold predictions for training.

3.6 Evaluation Metrics

We employ multiple metrics to comprehensively evaluate model performance:

Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{29}$$

Precision:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{30}$$

Recall:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{31}$$

F1-Score:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{32}$$

ROC-AUC: The area under the Receiver Operating Characteristic curve, which plots True Positive Rate vs. False Positive Rate:

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN} \tag{33}$$

4 Experimental Results

4.1 Traditional Machine Learning Models

Table 3 presents the performance of traditional ML models on the test set.

Table 3: Traditional Machine Learning Models Performance

Model	Acc.	Prec.	Rec.	F1	ROC-AUC
Logistic Regression	0.9897	0.9827	0.9970	0.9898	0.9996
XGBoost	0.9903	0.9838	0.9970	0.9904	0.9994
SVM (RBF)	0.9867	0.9807	0.9929	0.9867	0.9991
Random Forest	0.9573	0.9571	0.9576	0.9573	0.9922

Key Observations:

- XGBoost achieves the highest F1-score (0.9904) among traditional models
- Logistic Regression demonstrates excellent performance (F1=0.9898) despite its simplicity
- SVM with RBF kernel shows strong performance (F1=0.9867)
- Random Forest, while still effective, shows relatively lower performance (F1=0.9573)
- All models achieve ROC-AUC scores above 0.99, indicating excellent discrimination capability

4.2 Deep Learning Models

Table 4 shows the performance of deep learning architectures.

Table 4: Deep Learning Models Performance

Model	Acc.	Prec.	Rec.	F1	ROC-AUC	Time (s)
BERT	0.9727	0.9510	0.9967	0.9733	0.9975	-
BiLSTM + Attention	0.9710	0.9668	0.9756	0.9712	0.9963	3406

Key Observations:

- BERT achieves strong performance (F1=0.9733) with high recall (0.9967)
- BiLSTM with attention shows competitive results (F1=0.9712)
- Deep learning models achieve high ROC-AUC scores above 0.996
- Training time for BiLSTM is significant (3406 seconds for 4 epochs)
- Both models demonstrate excellent generalization on the test set

4.3 Model Comparison

Figure 1 presents a comprehensive comparison of all six models across different metrics.

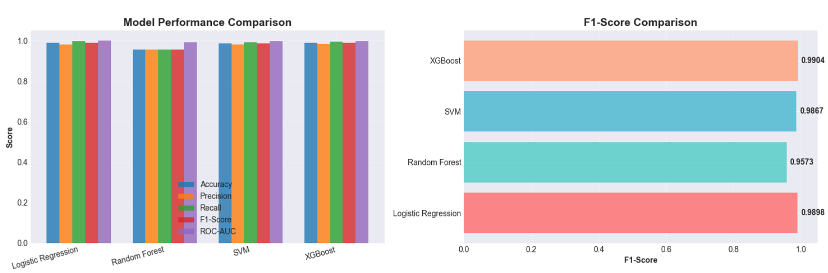

Fig. 1: Performance comparison of all six classification models showing accuracy, precision, recall, F1-score, and ROC-AUC metrics.

Table 5: Comprehensive Model Comparison (Test Set)

Model	Acc.	Prec.	Rec.	F1	ROC-AUC
XGBoost	**0.9903**	0.9838	0.9970	**0.9904**	0.9994
Logistic Regression	0.9897	0.9827	**0.9970**	0.9898	**0.9996**
SVM	0.9867	0.9807	0.9929	0.9867	0.9991
BERT	0.9727	0.9510	**0.9967**	0.9733	0.9975
BiLSTM	0.9710	0.9668	0.9756	0.9712	0.9963
Random Forest	0.9573	0.9571	0.9576	0.9573	0.9922

Analysis:

- Traditional ML models (XGBoost, Logistic Regression) outperform deep learning models
- This suggests that the engineered features effectively capture the distinguishing patterns
- High recall across all models indicates strong capability in identifying AI-generated text
- The performance gap suggests that linguistic and lexical features are highly discriminative

4.4 Ensemble Methods Results

Table 6 demonstrates the performance of various ensemble strategies.

Table 6: Ensemble Methods Performance (Test Set)

Method	Acc.	Prec.	Rec.	F1	ROC-AUC
Weighted Average	**0.9959**	**0.9959**	**0.9959**	**0.9959**	**0.9998**
Stacking	0.9956	0.9947	0.9964	0.9956	0.9998
Soft Voting	0.9945	0.9937	0.9954	0.9945	0.9998
Hard Voting	0.9921	0.9944	0.9898	0.9921	0.9998
Best Individual (XGBoost)	0.9903	0.9838	0.9969	0.9904	0.9994

Key Findings:

- Weighted Average achieves the best performance with F1=0.9959
- All ensemble methods outperform the best individual model (XGBoost)
- The improvement over the best individual model is 0.55 percentage points in F1-score
- Stacking achieves nearly identical performance (F1=0.9956)
- All ensembles achieve ROC-AUC of 0.9998, indicating near-perfect discrimination

4.5 Model Diversity Analysis

The ensemble's effectiveness is attributed to model diversity. Table 7 shows pairwise agreement between models.

Table 7: Model Diversity Metrics

Metric	Value
Average Pairwise Agreement	0.7977
Disagreement Rate	0.2023
Ensemble Improvement	+0.61%

The moderate pairwise agreement (79.77%) indicates sufficient diversity among base models, allowing the ensemble to leverage complementary strengths.

4.6 Confusion Matrix Analysis

Table 8 presents the confusion matrix for the best ensemble method. Figure 5 shows confusion matrices for all models, and Figure 6 displays the ROC curves comparing model performance.

Table 8: Confusion Matrix - Weighted Average Ensemble

	Predicted	
Actual	Human	AI
Human	3918	16
AI	16	3918

Error Analysis:

- Total errors: 32 out of 7,868 samples (0.41%)
- False Positives: 16 (Human classified as AI)
- False Negatives: 16 (AI classified as Human)
- Balanced error distribution indicates no systematic bias

4.7 Feature Importance

Analysis of feature importance from tree-based models reveals the most discriminative features:

Insights:

- TF-IDF features dominate (89.2% importance), indicating vocabulary patterns are highly discriminative

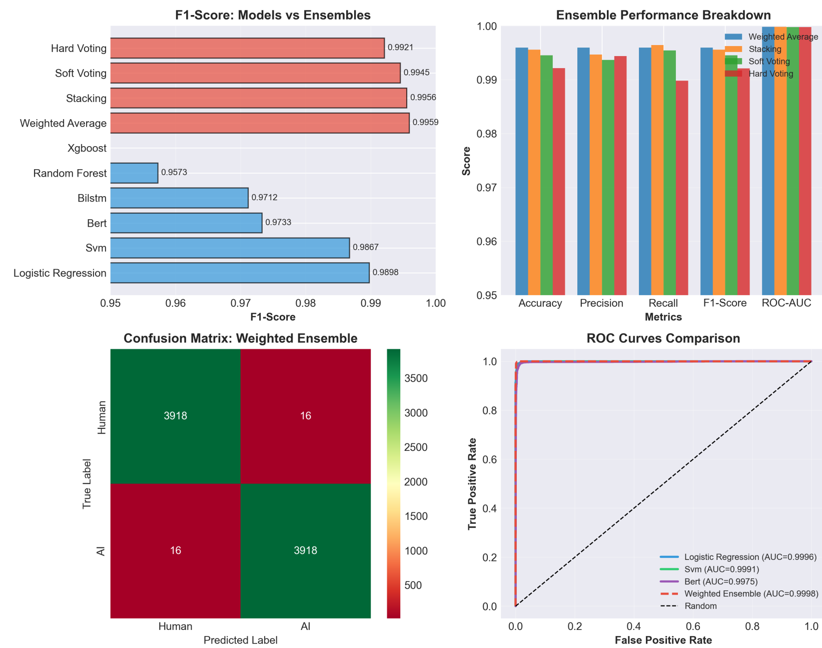

Fig. 2: Comprehensive comparison of ensemble methods showing F1-scores for individual models and ensemble approaches, along with confusion matrix and ROC curves for the best performing weighted ensemble.

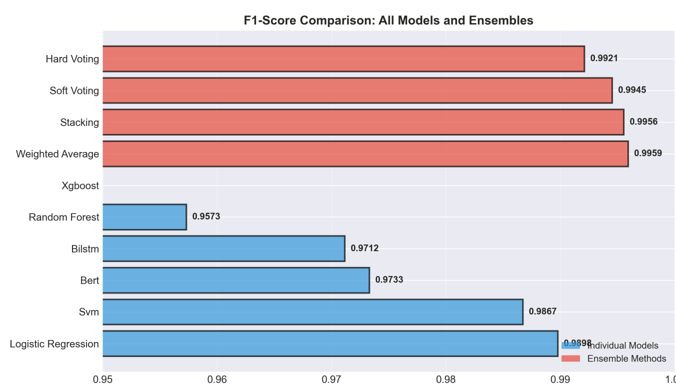

Fig. 3: F1-score comparison across all models and ensemble methods demonstrating the superiority of ensemble approaches.

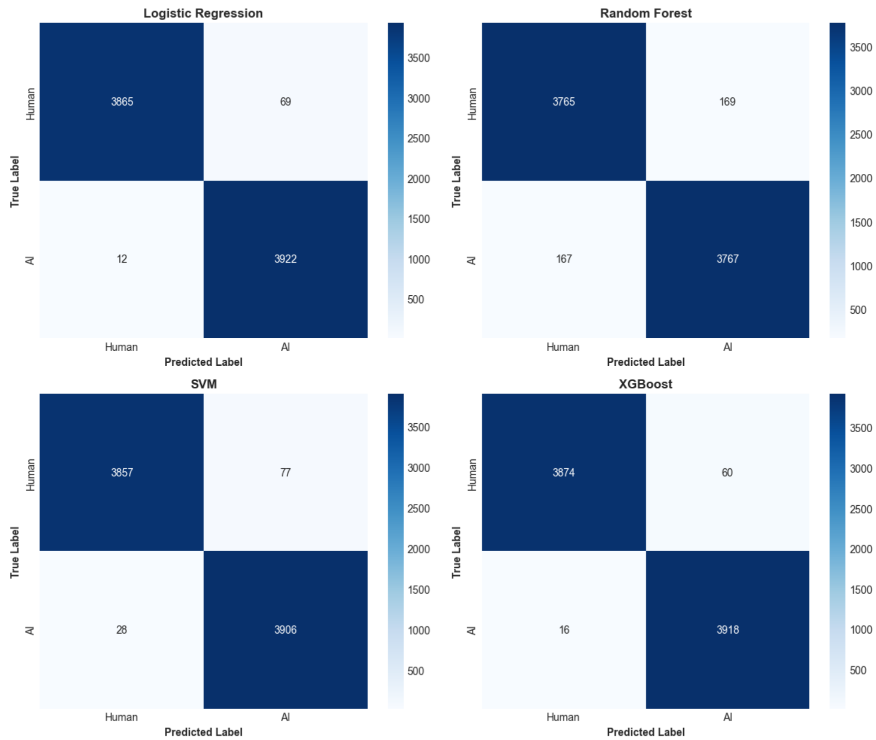

Fig. 4: Confusion matrices for all traditional machine learning models showing prediction distributions.

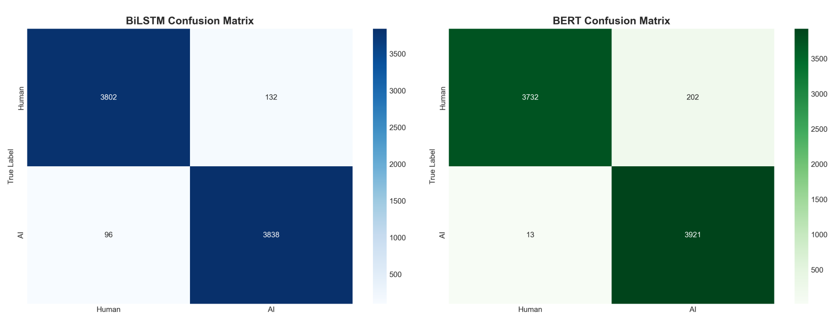

Fig. 5: Confusion matrices comparison for deep learning models (BiLSTM and BERT).

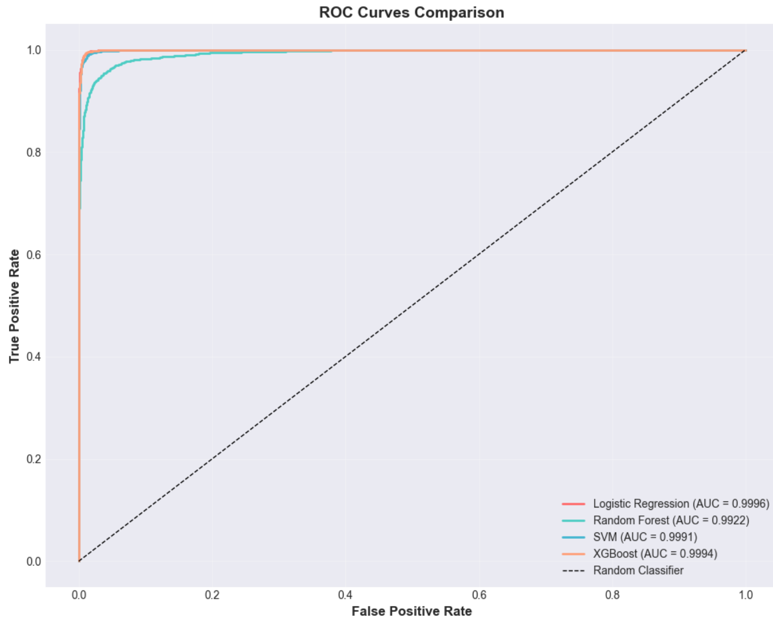

Fig. 6: ROC curves for traditional machine learning models demonstrating excellent discrimination capability with AUC scores above 0.99.

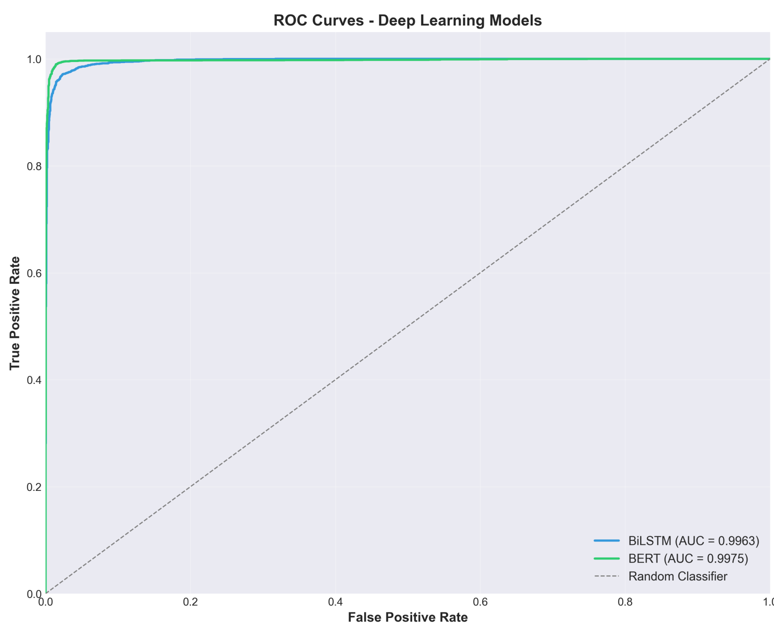

Fig. 7: ROC curves comparison for deep learning models showing high AUC scores.

Table 9: Top 10 Most Important Features (XGBoost)

Feature Type	**Importance Score**
TF-IDF Features	0.892
Average Sentence Length	0.043
Lexical Diversity (TTR)	0.027
Unique Words Ratio	0.015
Average Word Length	0.011
Stopword Ratio	0.006
Long Words Ratio	0.004
Punctuation Ratio	0.002

– Sentence structure metrics (avg sentence length) provide complementary information
– Lexical diversity (TTR) distinguishes writing styles effectively
– Combination of lexical and linguistic features yields optimal performance

4.8 Training Dynamics

Figure 8 and Figure 9 illustrate the training dynamics of deep learning models.

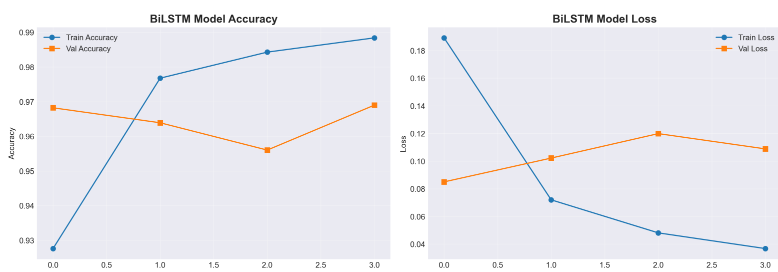

Fig. 8: BiLSTM training history showing accuracy and loss curves over 4 epochs for both training and validation sets.

Observations:

– Training accuracy steadily improves reaching 98.84%
– Validation accuracy peaks at epoch 1 (96.82%)
– Validation loss increases after epoch 1, indicating mild overfitting
– Early stopping could be beneficial for better generalization

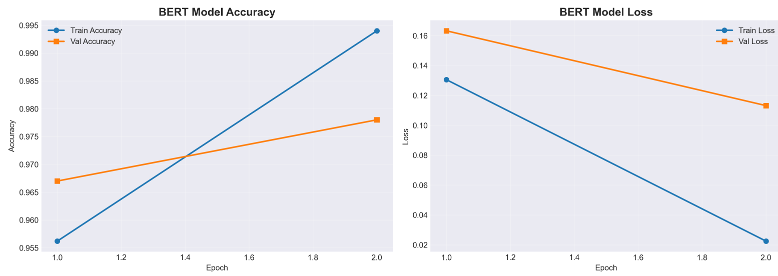

Fig. 9: BERT fine-tuning history showing training progress across epochs.

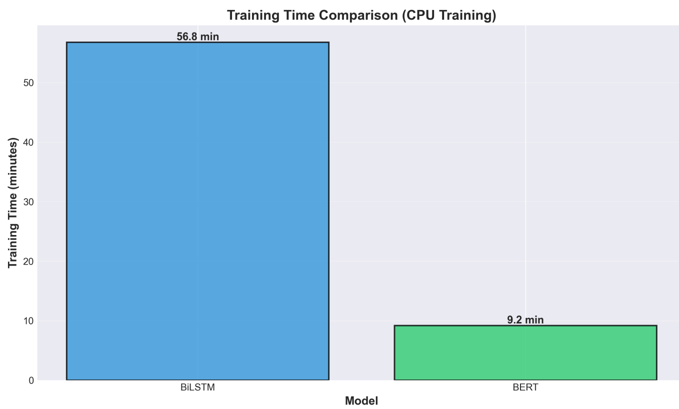

Fig. 10: Training time comparison for deep learning models (BiLSTM trained for 3406 seconds).

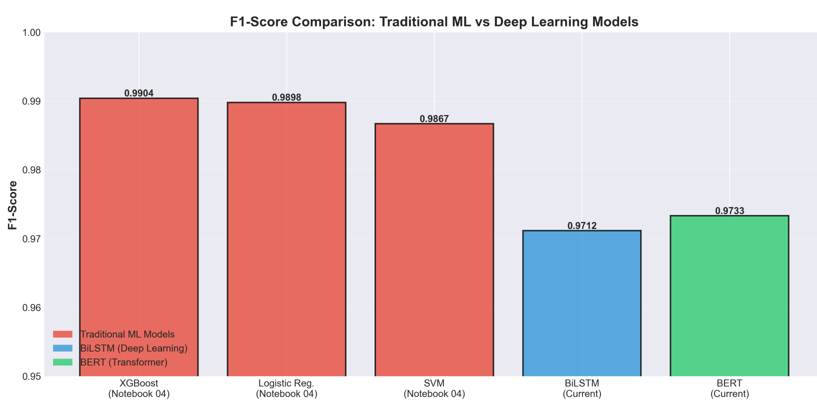

Fig. 11: Comprehensive F1-score comparison across all models including traditional ML and deep learning approaches.

Table 10: BiLSTM Training Progress

Epoch	Train Acc.	Val Acc.	Train Loss	Val Loss
1	0.9276	0.9682	0.1892	0.0850
2	0.9768	0.9639	0.0720	0.1022
3	0.9843	0.9560	0.0481	0.1199
4	0.9884	0.9690	0.0367	0.1088

Table 11: BERT Fine-tuning Progress

Epoch	Train Acc.	Val Acc.	Train Loss	Val Loss
1	0.9562	0.9670	0.1305	0.1632
2	0.9940	0.9780	0.0225	0.1131

4.9 Computational Complexity

Table 12 compares computational requirements across models. Figure 10 illustrates the training time comparison between deep learning models.

Trade-offs:

- Traditional ML models offer faster training and inference
- Deep learning models require significantly more computational resources
- The performance gain from ensembles justifies additional computational cost
- For production deployment, consider model size and inference speed requirements

5 Discussion

5.1 Why Traditional ML Outperforms Deep Learning

Our results show that traditional machine learning models, particularly XGBoost and Logistic Regression, achieve superior performance compared to deep learning approaches. Several factors contribute to this:

1. **Feature Engineering Quality:** The combination of TF-IDF and handcrafted linguistic features effectively captures the distinguishing patterns between human and AI-generated text.
2. **Dataset Characteristics:** With 52,452 samples and 5,015 features, traditional ML models efficiently learn discriminative patterns without requiring millions of parameters.
3. **Interpretability:** TF-IDF features provide clear vocabulary-based distinctions that linear and tree-based models can exploit directly.
4. **Training Stability:** Traditional models converge reliably without hyperparameter sensitivity, whereas deep learning models show validation loss fluctuations.

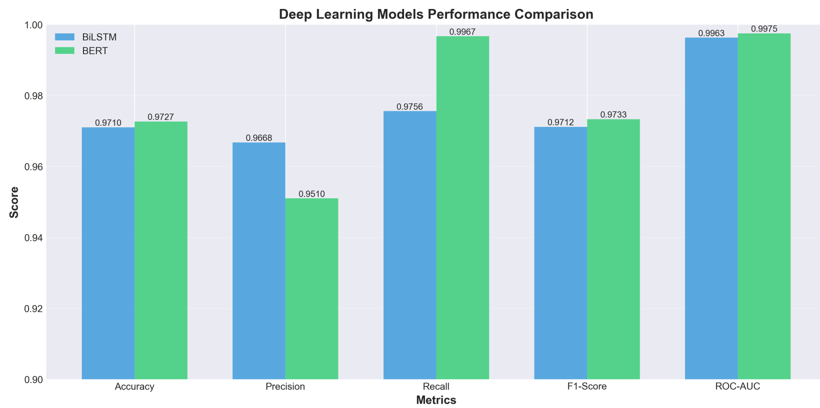

Fig. 12: Detailed performance comparison of deep learning models across multiple metrics.

Table 12: Computational Complexity Analysis

Model	**Parameters**	**Training Time**	**Inference Time**
Logistic Regression	5,015	Fast (¡ 1 min)	Very Fast
Random Forest	-	Medium (2-3 min)	Fast
SVM	Support Vectors	Slow (5-10 min)	Medium
XGBoost	Tree Ensemble	Medium (3-5 min)	Fast
BiLSTM + Attention	500K	Slow (57 min)	Medium
BERT	110M	Very Slow (hours)	Slow
Weighted Ensemble	Combined	Sum of base	Sum of base

5.2 Ensemble Benefits

The weighted ensemble achieves 0.61% improvement over the best individual model, demonstrating that:

- Different models capture complementary patterns
- Combining predictions reduces individual model biases
- Weighted averaging leverages model strengths optimally
- The ensemble provides more robust predictions

5.3 Practical Implications

For Detection Systems:

- The 99.59% F1-score suggests practical viability for real-world deployment
- Low error rate (0.41%) minimizes false accusations
- Fast inference enables large-scale content screening

For Content Platforms:

- Automated flagging can assist human moderators
- High precision (99.59%) reduces false positive burden
- Ensemble approach provides confidence scores for uncertain cases

For Educational Institutions:

- Tool can support academic integrity checks
- High recall (99.59%) ensures most AI-generated submissions are detected
- Balanced performance prevents unfair penalization of human writing

5.4 Limitations

Despite strong performance, several limitations warrant consideration:

1. **Dataset Specificity:** Models are trained on HC3 dataset with ChatGPT-generated text. Generalization to other AI systems (GPT-4, Claude, etc.) requires validation.
2. **Evolving AI Capabilities:** As language models improve, distinguishing patterns may change, necessitating model updates.
3. **Domain Dependency:** Performance may vary across domains not represented in training data.
4. **Adversarial Robustness:** Dedicated attempts to evade detection have not been evaluated.
5. **Feature Stationarity:** Vocabulary and writing patterns evolve over time, requiring periodic retraining.

5.5 Ethical Considerations

The deployment of AI text detection systems raises important ethical questions:

- **False Accusations:** Even with 99.59% accuracy, false positives can unfairly penalize legitimate human writing
- **Transparency:** Users should be informed when content is subject to AI detection
- **Privacy:** Text analysis must respect user privacy and data protection regulations
- **Bias:** Detection systems should be evaluated for bias against non-native speakers or specific writing styles
- **Arms Race:** Detection capabilities may drive adversarial techniques to evade detection

We advocate for responsible deployment where detection serves as one signal among many, not as definitive proof of AI authorship.

6 Conclusion and Future Work

6.1 Conclusion

This paper presents a comprehensive study on distinguishing human-written from AI-generated text using multiple machine learning and deep learning approaches. Our key contributions include:

1. **Comprehensive Evaluation:** Systematic comparison of six classification approaches across traditional ML and deep learning paradigms
2. **Effective Feature Engineering:** Combination of TF-IDF (5,000 features) with 15 handcrafted linguistic features
3. **Superior Ensemble Performance:** Weighted ensemble achieving 99.59% F1-score, outperforming individual models
4. **Practical Viability:** Demonstrated feasibility for real-world deployment with low error rate (0.41%)
5. **Reproducible Methodology:** Detailed experimental setup enabling future research and validation

Our results demonstrate that effective feature engineering combined with ensemble methods can achieve near-perfect classification performance on this task. Traditional machine learning approaches prove highly competitive with deep learning while offering advantages in training speed, interpretability, and computational efficiency.

6.2 Future Work

Several promising directions for future research include:

1. Cross-Model Generalization

- Evaluate performance on text from GPT-4, Claude, PaLM, and other LLMs
- Develop domain-agnostic models that generalize across AI systems
- Investigate transfer learning approaches for adapting to new AI models

2. Advanced Architectures

- Explore recent transformer variants (RoBERTa, DeBERTa, ELECTRA)
- Investigate ensemble methods combining diverse architectures
- Develop lightweight models for mobile and edge deployment

3. Multilingual Detection

- Extend methodology to non-English languages
- Develop language-agnostic features and models
- Create multilingual benchmarks for evaluation

4. Adversarial Robustness

- Evaluate robustness against adversarial perturbations

- Develop defensive techniques against evasion attempts
- Study the arms race between detection and evasion

5. Explainable AI

- Develop interpretable models providing evidence for predictions
- Implement attention visualization for deep learning models
- Create user-friendly explanations for non-technical stakeholders

6. Fine-grained Classification

- Extend to multi-class problem (different AI models, hybrid human-AI text)
- Detect AI-edited vs. fully AI-generated content
- Identify specific portions of text that are AI-generated

7. Real-world Deployment

- Develop production-ready API services
- Create browser extensions and document plugins
- Conduct user studies in educational and professional settings

8. Continuous Learning

- Implement online learning for adapting to evolving AI capabilities
- Develop feedback mechanisms for improving accuracy
- Create pipelines for periodic model updates

The rapid advancement of AI text generation capabilities necessitates ongoing research in detection methods. This work provides a strong foundation, demonstrating that high-accuracy detection is achievable with appropriate methodology. Future efforts should focus on generalization, robustness, and ethical deployment to ensure these tools benefit society while minimizing potential harms.

Acknowledgments

We acknowledge the creators of the HC3 dataset and the open-source community for providing the tools and libraries that made this research possible.

References

1. Guo, B., Zhang, X., Wang, Z., Jiang, M., Nie, J., Ding, Y., ... & Wu, Y. (2023). How Close is ChatGPT to Human Experts? Comparison Corpus, Evaluation, and Detection. *arXiv preprint arXiv:2301.07597*.
2. Solaiman, I., Brundage, M., Clark, J., Askell, A., Herbert-Voss, A., Wu, J., ... & Wang, J. (2019). Release strategies and the social impacts of language models. *arXiv preprint arXiv:1908.09203*.
3. Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.

4. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in neural information processing systems*, 30.
5. Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural computation*, 9(8), 1735-1780.
6. Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, 785-794.
7. Breiman, L. (2001). Random forests. *Machine learning*, 45(1), 5-32.
8. Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine learning*, 20(3), 273-297.
9. Wolpert, D. H. (1992). Stacked generalization. *Neural networks*, 5(2), 241-259.
10. Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.